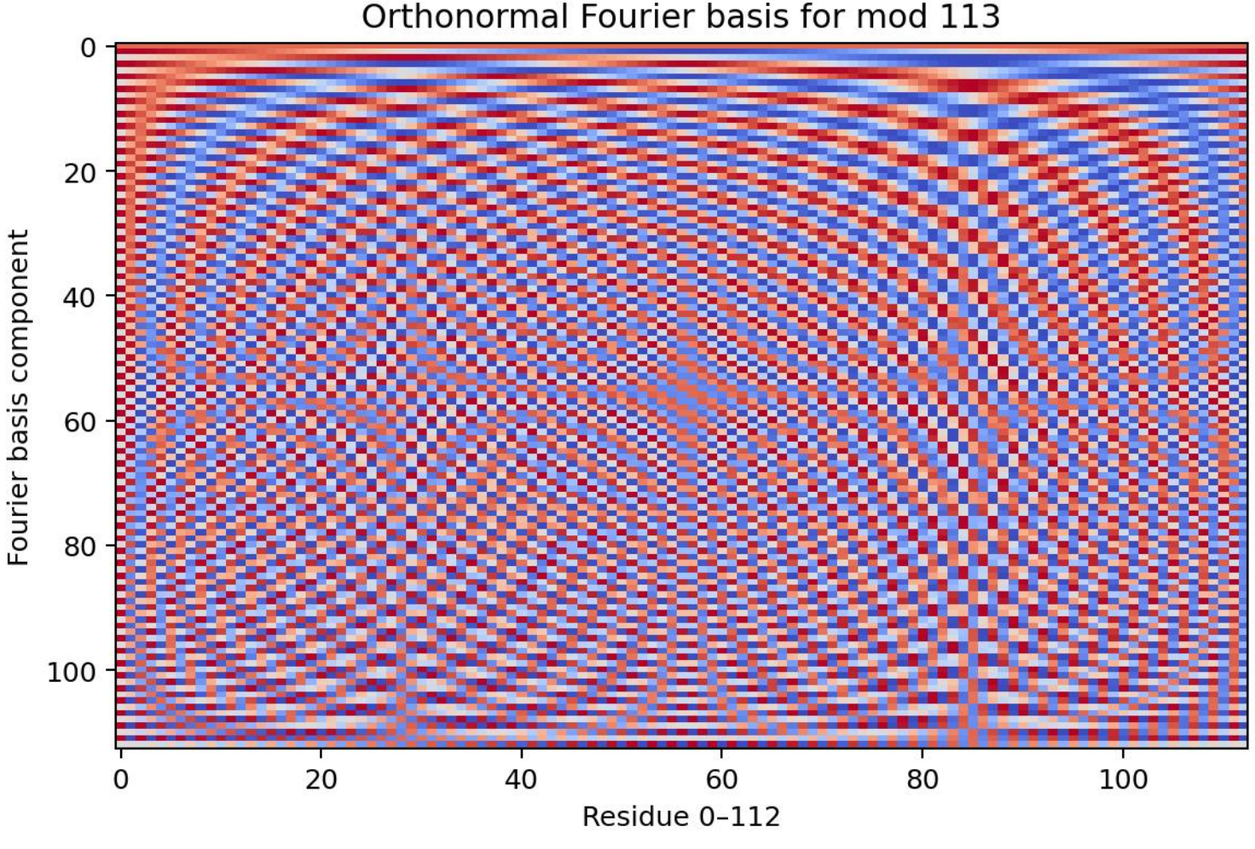
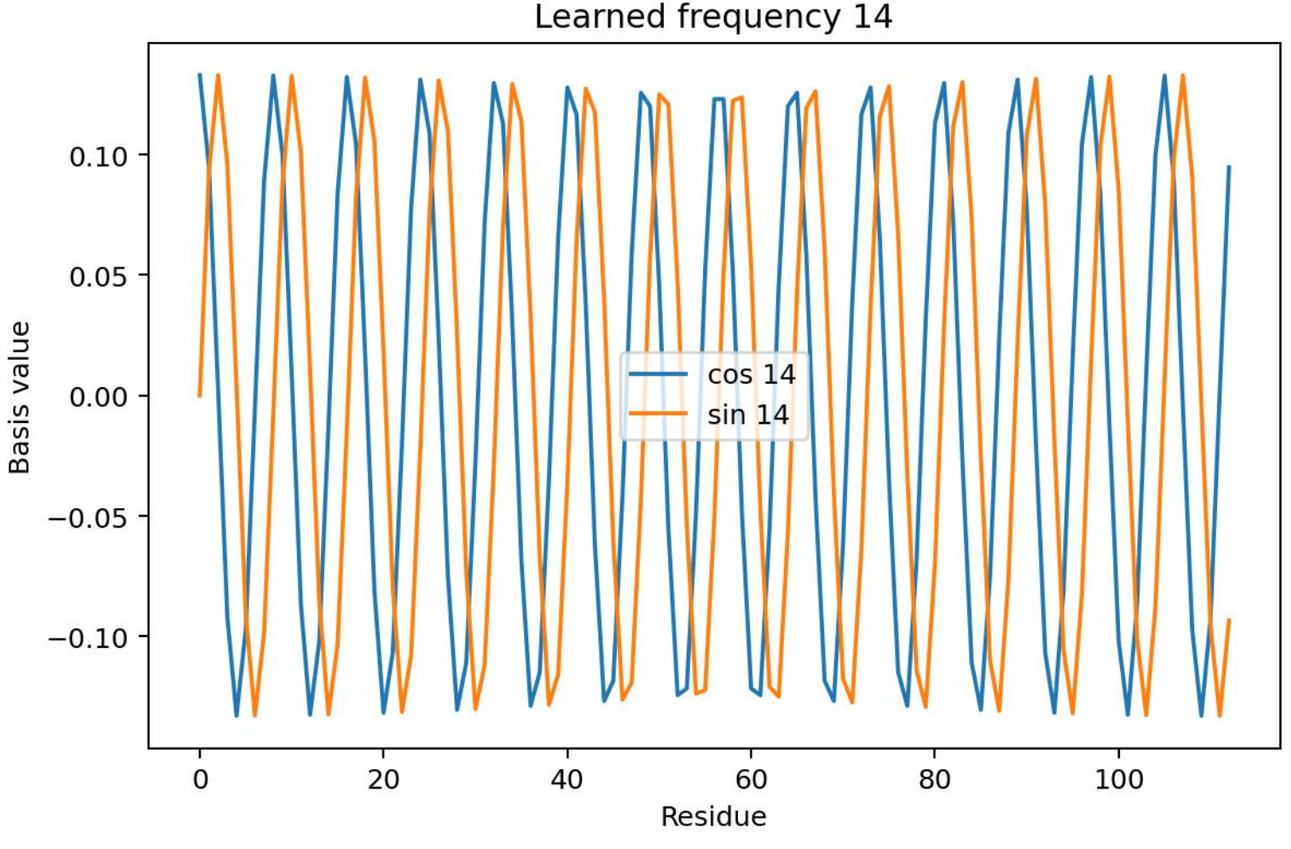
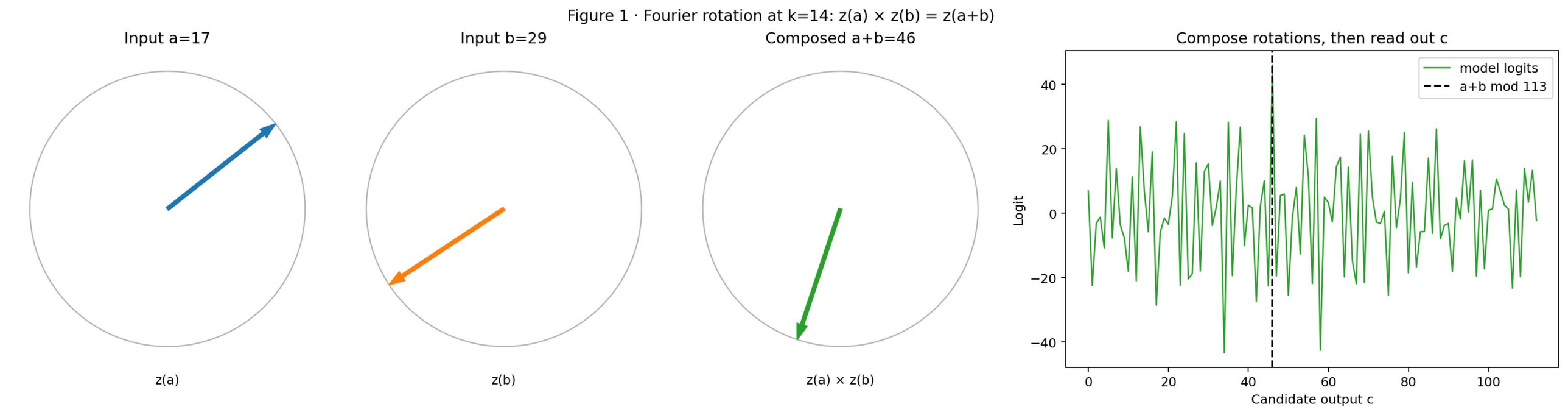

Orthonormal Fourier basis for mod 113
Fourier basis component
Residue 0–112
0
20
40
60
80
100


Learned frequency 14
Basis value
Residue
cos 14
sin 14
0.10
0.05
0.00
−0.05
−0.10
0
20
40
60
80
100


Figure 1 · Fourier rotation at k=14: z(a) × z(b) = z(a+b)
Input a=17
Input b=29
Composed a+b=46
z(a)
z(b)
z(a) × z(b)
Compose rotations, then read out c
model logits
a+b mod 113
Logit
Candidate output c
40
20
0
−20
−40
0
20
40
60
80
100

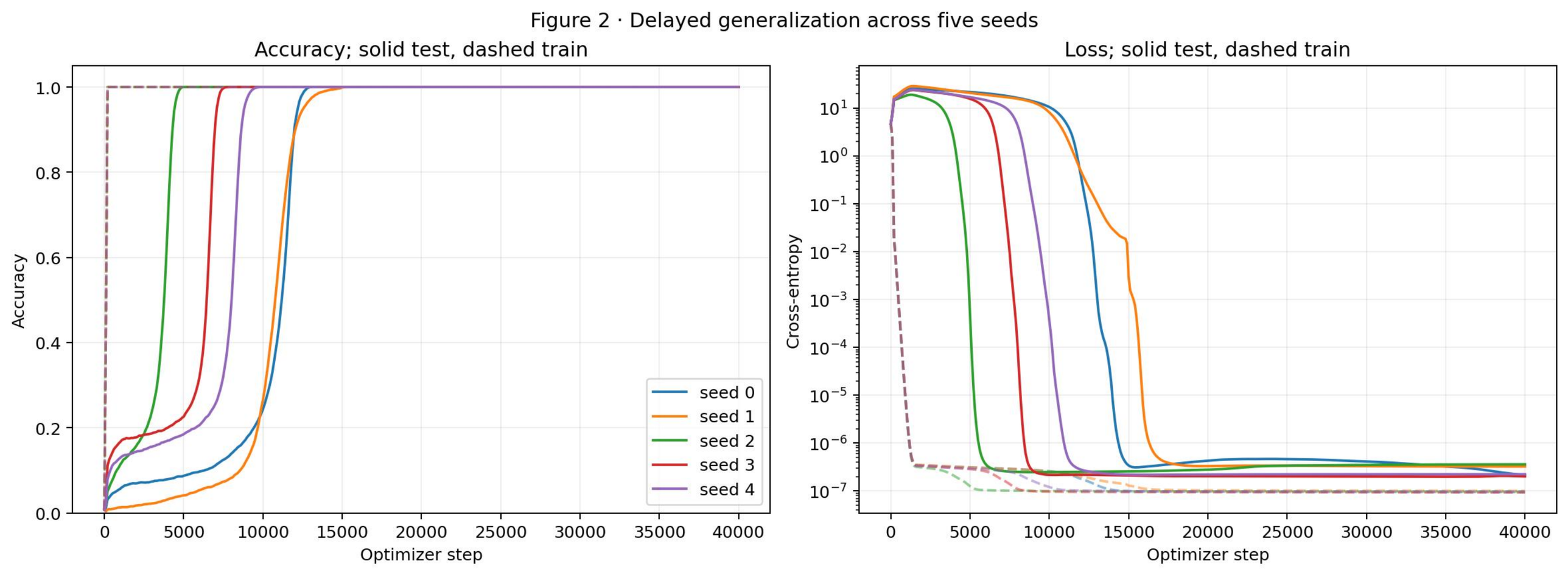
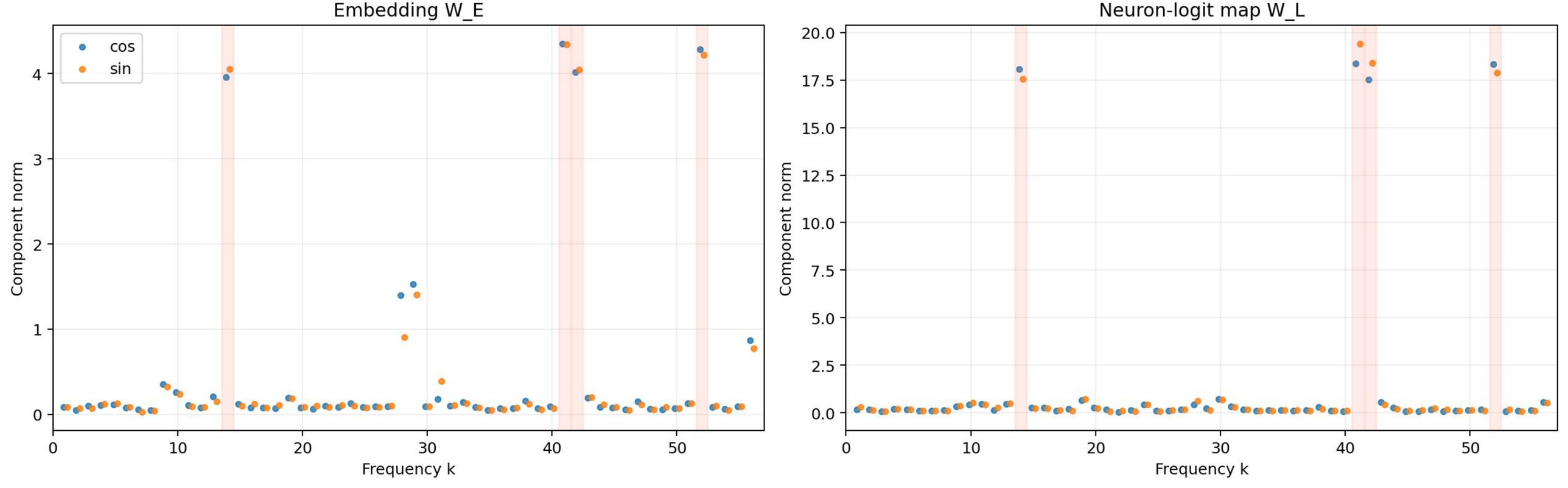

Figure 2 · Delayed generalization across five seeds
Accuracy; solid test, dashed train
Loss; solid test, dashed train
Accuracy
Cross-entropy
Optimizer step
Optimizer step
seed 0
seed 1
seed 2
seed 3
seed 4

Figure 3 · Learned weights in the Fourier basis



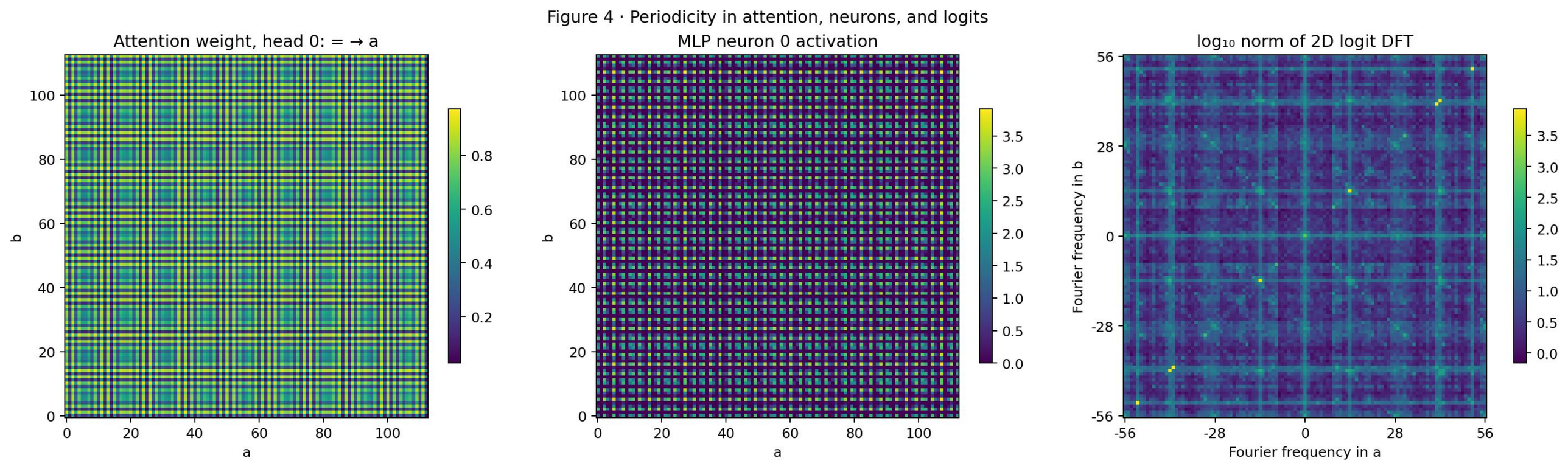
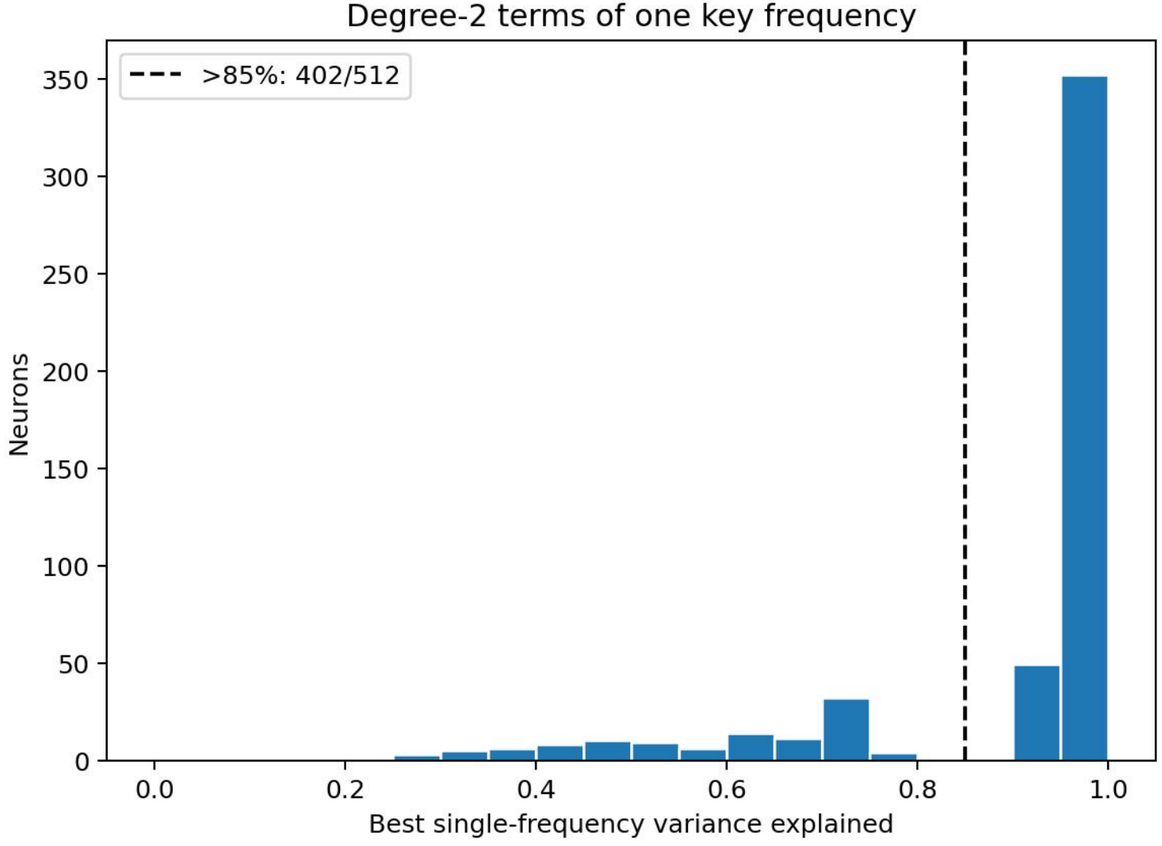
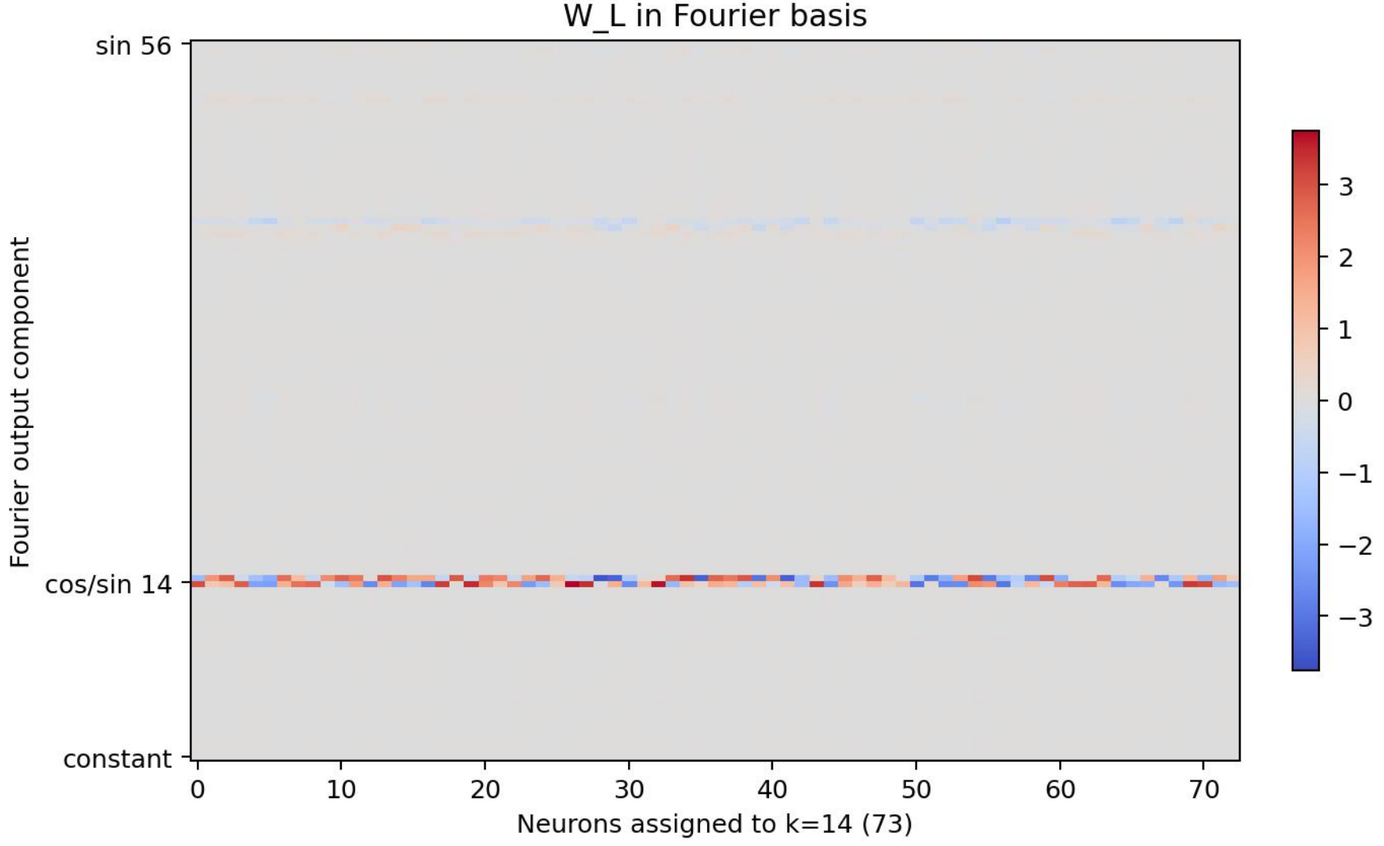

Figure 4 · Periodicity in attention, neurons, and logits
Attention weight, head 0: = → a
MLP neuron 0 activation
log₁₀ norm of 2D logit DFT
b
a
Fourier frequency in b
Fourier frequency in a

Figure 5 · Neuron frequency specialization and logit map





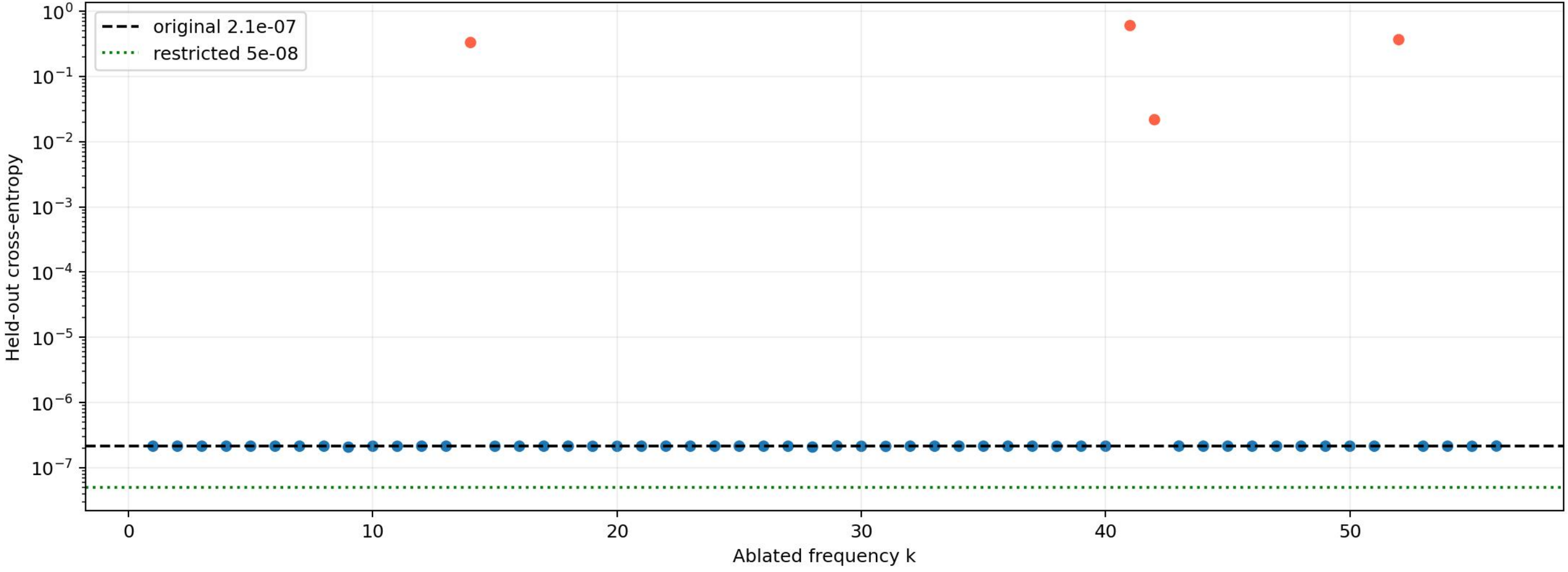

Figure 6 · Remove one cos/sin(k(a+b)) component
original 2.1e-07
restricted 5e-08
Held-out cross-entropy
Ablated frequency k
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
0
10
20
30
40
50

Figure 7 · Progress measures through grokking

